# EXPERIMENT-3

**Student Name:** Ridhima Gulati
**UID:** 24BCY70133
**Branch:** AIT-CSE (Cyber Security)
**Section/Group:** 24AIT_KRG-G2
**Semester:** 5
**Date:** 29/07/2026
**Subject Name**: Computer Networks
**Subject Code:** 24CSH-337

## AIM:

To understand the concepts of:

1. Hub
2. Layer-2 Switch
3. Router
4. Difference between Hub and Switch

and simulate their working using Cisco Packet Tracer.

## OBJECTIVES:

- To study the working of a Hub.
- To understand data transmission through broadcasting.
- To study the working of a Layer-2 Switch.
- To understand MAC Address Table learning.
- To study the role of a Router in connecting multiple LANs.
- To configure IP Addressing and Default Gateway.
- To verify communication between devices using Simple PDU and Ping.
- To compare Hub and Switch based on functionality and performance.

## SOFTWARE REQUIREMENTS:

**(a) Software:**

- Cisco Packet Tracer
- Windows Operating System

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**(b) Hardware (Virtual):**

- PC-PT
- Hub-PT
- Cisco 2960-24TT Switch
- Cisco 2911 Router

**THEORY:** (i) Hub

A Hub is a networking device used to connect multiple computers in a Local Area Network (LAN). It works at the Physical Layer (Layer 1) of the OSI model and simply forwards electrical signals without examining any address information.

A hub has multiple ports and is commonly used to create a Star Topology. Whenever a packet arrives on one port, the hub copies the packet and broadcasts it to every other connected port. Every device receives the frame, but only the intended destination accepts it while the others discard it.

Since the hub broadcasts all traffic, it creates unnecessary network congestion and increases collisions. It operates only in Half-Duplex mode and does not maintain any memory or MAC address table.

**Advantages**

- Less expensive than switches.
- Easy to install.
- Suitable for small LANs.

**Disadvantages**

- Broadcasts all packets.
- No memory.
- High collision domain.
- Half-duplex communication.
- Low security.

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(ii) Layer-2 Switch

A Switch is a Layer-2 networking device that connects devices within a LAN. Unlike a hub, a switch maintains a MAC Address Table in its memory.

Whenever a frame arrives, the switch learns the source MAC address and stores it along with the port number. Later, when another frame is sent, the switch forwards it only to the destination port instead of broadcasting to all devices.

This intelligent forwarding minimizes collisions, improves bandwidth utilization, and provides better security and performance.

**Features**

- Layer-2 device.
- Stores MAC Address Table.
- Intelligent forwarding.
- Supports Unicast, Broadcast and Multicast.
- Full-Duplex communication.
- More efficient than Hub.

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(iii) Router

A Router is a Layer-3 networking device used to connect two or more different networks. It forwards packets based on IP Addresses.

A router maintains a Routing Table that stores the best path to different networks. Each router interface belongs to a different network.

In this experiment, Router 2911 connects:

- LAN-1 → 10.0.0.0
- LAN-2 → 192.168.1.0

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The router interfaces act as the Default Gateway for their respective LANs, enabling communication between devices on different networks.

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Difference between Hub, Switch and Router:

| Feature | Hub | Switch | Router |
|---|---|---|---|
| **OSI Layer** | Layer 1 (Physical) | Layer 2 (Data Link) | Layer 3 (Network) |
| **Main Function** | Connect devices in LAN | Connect devices intelligently in LAN | Connect different networks |
| **Uses** | Electrical signals | MAC Addresses | IP Addresses |
| **Address Table** | No | MAC Address Table | Routing Table |
| **Memory** | No Memory | Has Memory | Has Memory |
| **Intelligence** | Not Intelligent | Intelligent | Highly Intelligent |
| **Data Forwarding** | Broadcasts to all ports | Sends only to destination MAC | Sends to destination network |
| **Collision Domain** | One | Separate for every port | Separate for every interface |
| **Broadcast Domain** | One | One (without VLAN) | Separate for each interface |
| **Duplex Mode** | Half Duplex | Full Duplex | Full Duplex |
| **Bandwidth Sharing** | Shared | Dedicated | Dedicated |
| **Security** | Low | Medium | High |
| **Speed** | Slow | Fast | Moderate (routing based) |
| **Cost** | Lowest | Medium | Highest |

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| | | | |
|---|---|---|---|
| **Configuration** | Not Required | Minimal | Complex |
| **Can connect different LANs?** | No | No | Yes |
| **Used as Default Gateway** | No | No | Yes |
| **Routing Capability** | No | No | Yes |
| **Suitable For** | Small LANs, Labs | Office/College LANs | Enterprise Networks, Internet |

**STEPS/IMPLEMENTATION:**

Part I – Hub

1. Add one Hub.
2. Add six PCs.
3. Connect all PCs using Copper Straight-Through Cable.
4. Assign IP addresses:

| PC | IP |
|---|---|
| PC0 | 10.10.10.1 |
| PC1 | 10.10.10.2 |
| PC2 | 10.10.10.3 |
| PC3 | 10.10.10.4 |
| PC4 | 10.10.10.5 |
| PC5 | 10.10.10.6 |

5. Switch to Simulation Mode.
6. Send Simple PDU from 10.10.10.1 to 10.10.10.5.
7. Observe packet broadcasting.

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Part II – Switch

1. Add one Cisco 2960 Switch.
2. Add four PCs.
3. Connect using Straight-Through cables.
4. Observe MAC Address learning.
5. Verify communication using Ping.
6. Observe that packets are forwarded only to the destination port.

Part III – Router

**1. Create**
**LAN-1** Network:
10.0.0.0
Subnet Mask:
255.0.0.0
Assign

PC0 → 10.0.0.1

PC1 → 10.0.0.2

PC2 → 10.0.0.3

Router G0/0 10.0.0.4
**2. Create**
**LAN-2** Network
192.168.1.0
Subnet Mask

255.255.255.0

Assign

PC3 → 192.168.1.1

PC4 → 192.168.1.2

PC5 → 192.168.1.3

Router G0/1
192.168.1.4

**3. Configure Default Gateway**

LAN-1
10.0.0.4
LAN-2
192.168.1.4
Verify communication using Ping.

**PROCEDURE:**

1. Hub

- Place Hub and six PCs.
- Connect using Straight-through cables.
- Assign IP addresses.
- Send Simple PDU.
- Observe broadcasting.

2. Switch

- Place one Switch.

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- Connect four PCs.
- Assign IP addresses.
- Verify MAC learning.
- Ping between PCs.

3. Router

- Create two LANs.
- Connect each LAN to Router.
- Configure router interfaces.
- Configure Default Gateway.
- Verify communication between LANs using Ping.
- 4. Layer-2 Switch

**INPUT/OUTPUT DETAILS:**

(a) Devices Used

- Hub-PT
- Cisco 2960 Switch
- Cisco 2911 Router
- PC-PT

(b) Cable Types

- Copper Straight-Through

(c) Router Interface Configuration

| Interface | IP Address |
|---|---|
| GigabitEthernet0/0 | 10.0.0.4 |
| GigabitEthernet0/1 | 192.168.1.4 |

(d) Network Details

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**LAN-1**

Network Address
10.0.0.0
Mask
255.0.0.0
Gateway
10.0.0.4

**LAN-2**

Network Address
192.168.1.0
Mask 255.255.255.0
Gateway 192.168.1.4

**DEMONSTRATION/OUTPUT:**

- The Hub broadcasts packets to all connected devices.
- The Switch forwards packets only to the required destination using its MAC Address Table.
- The Router successfully transfers packets between two different LANs using routing and default gateway configuration.
- Communication between devices was verified successfully using Ping and Simple PDU.

Part-1: Hub

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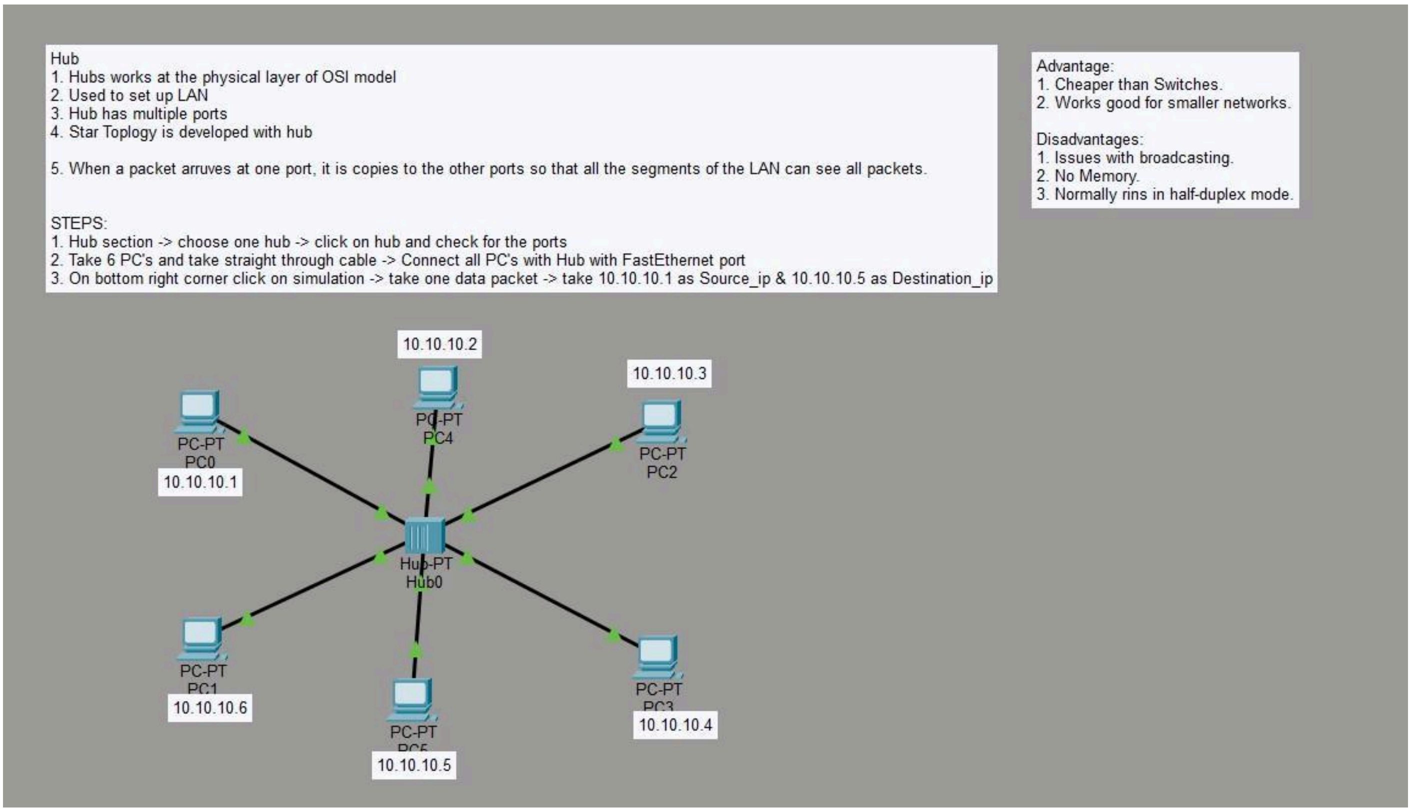


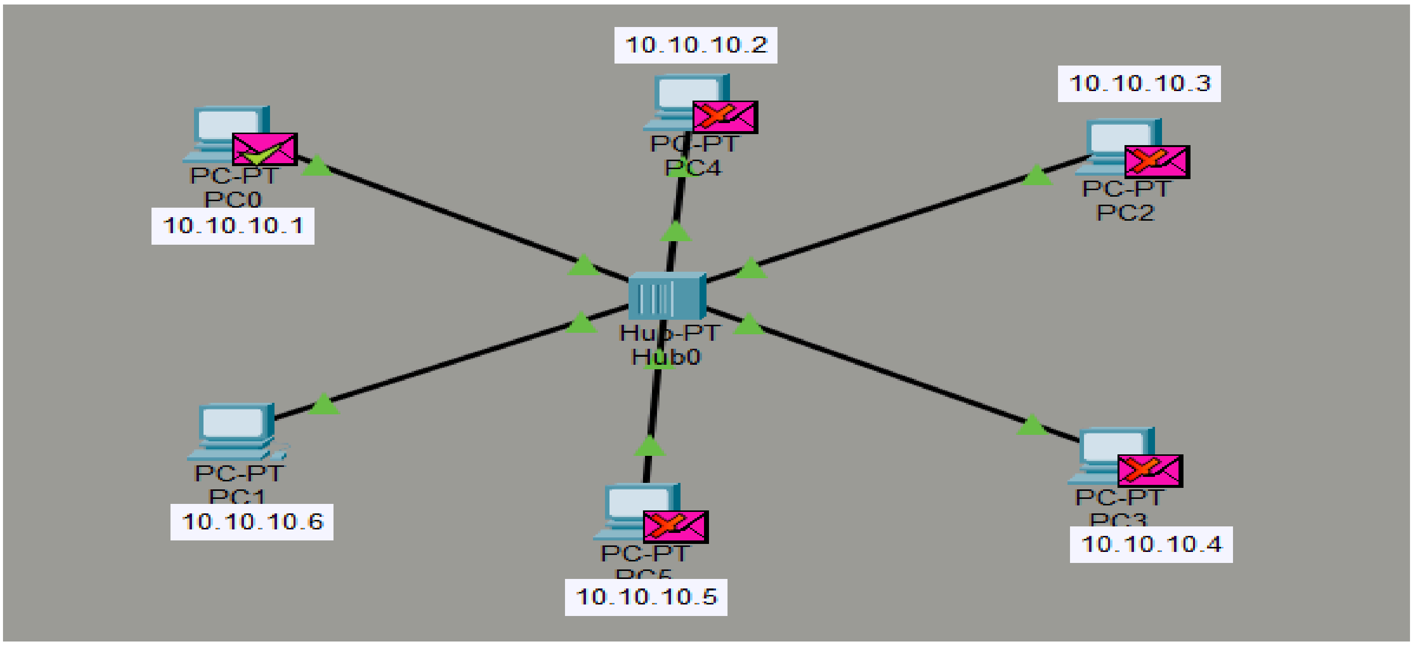


Part-2: Switch

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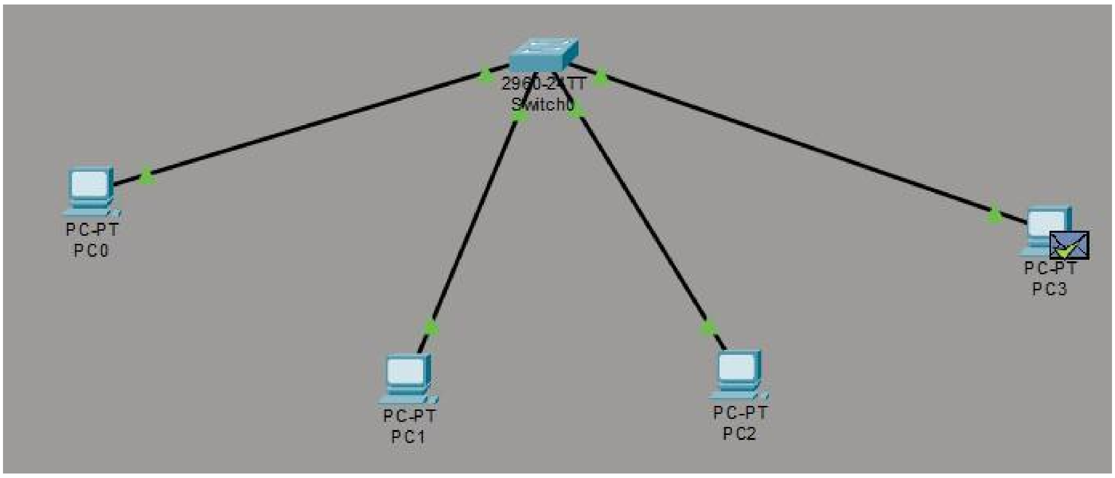
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time=1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.3

Pinging 10.10.10.3 with 32 bytes of data:

Reply from 10.10.10.3: bytes=32 time<1ms TTL=128
Reply from 10.10.10.3: bytes=32 time<1ms TTL=128
Reply from 10.10.10.3: bytes=32 time<1ms TTL=128
Reply from 10.10.10.3: bytes=32 time=1ms TTL=128

Ping statistics for 10.10.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```



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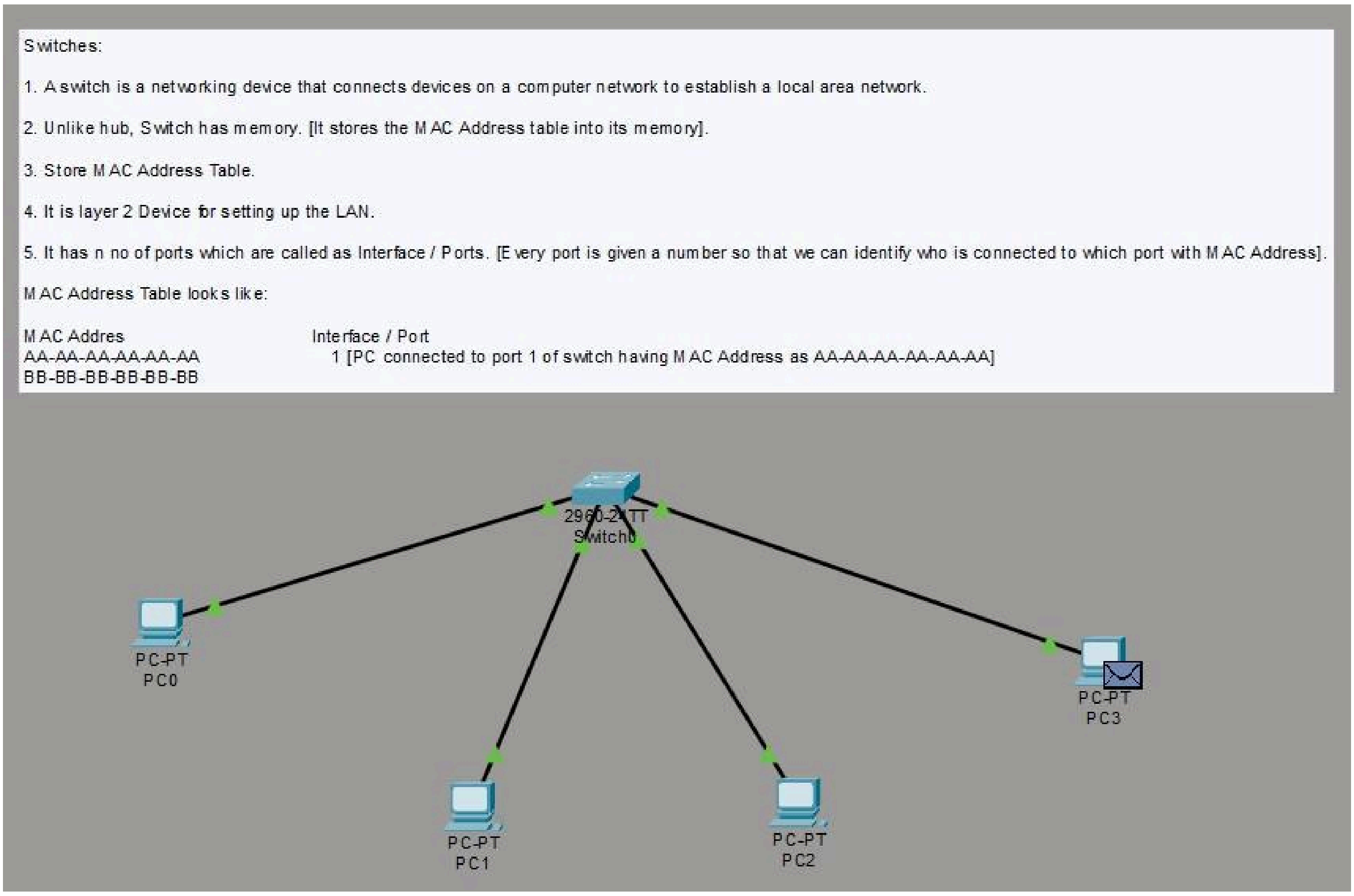
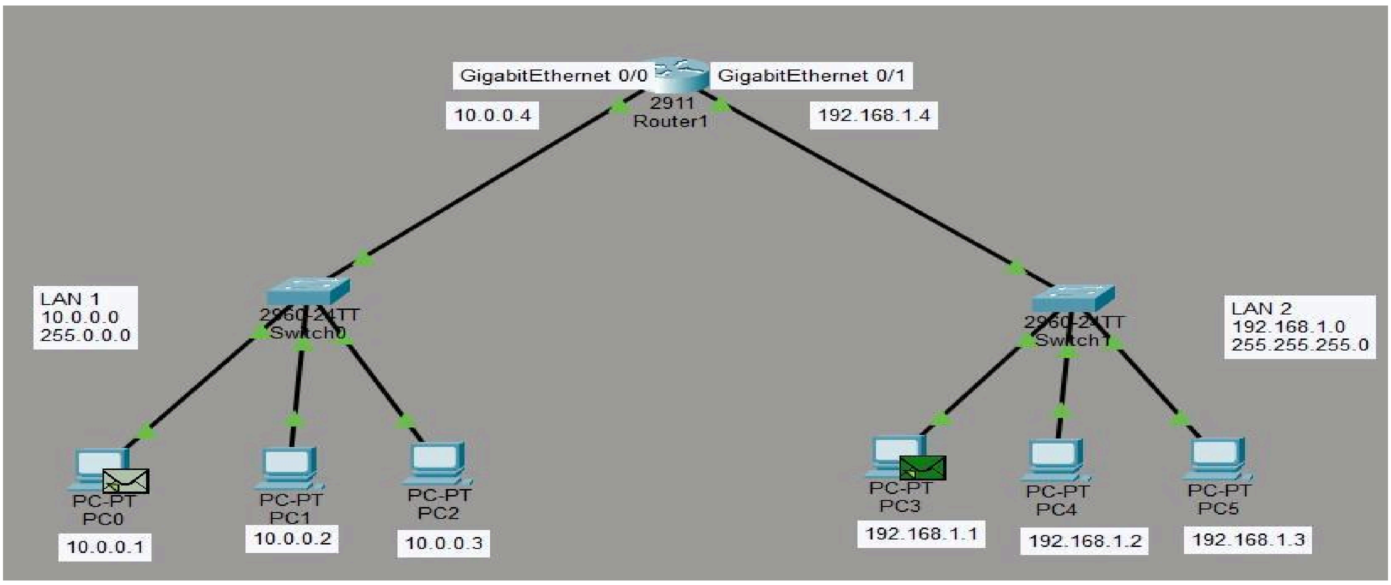



Part-3: Router



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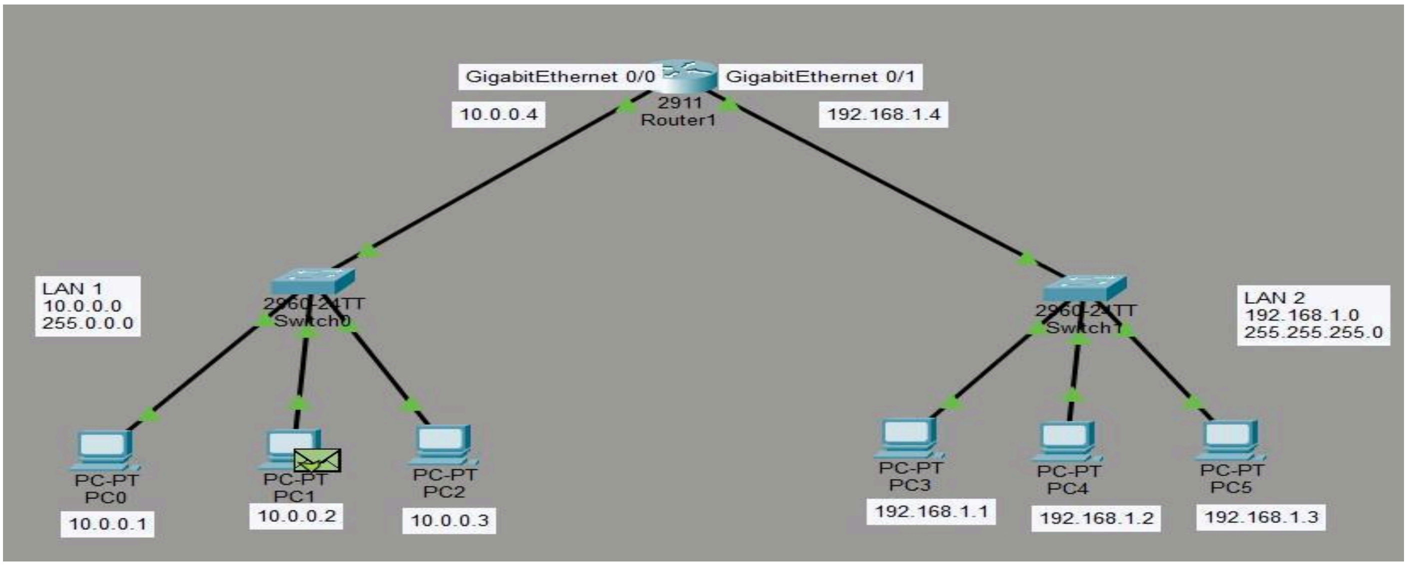


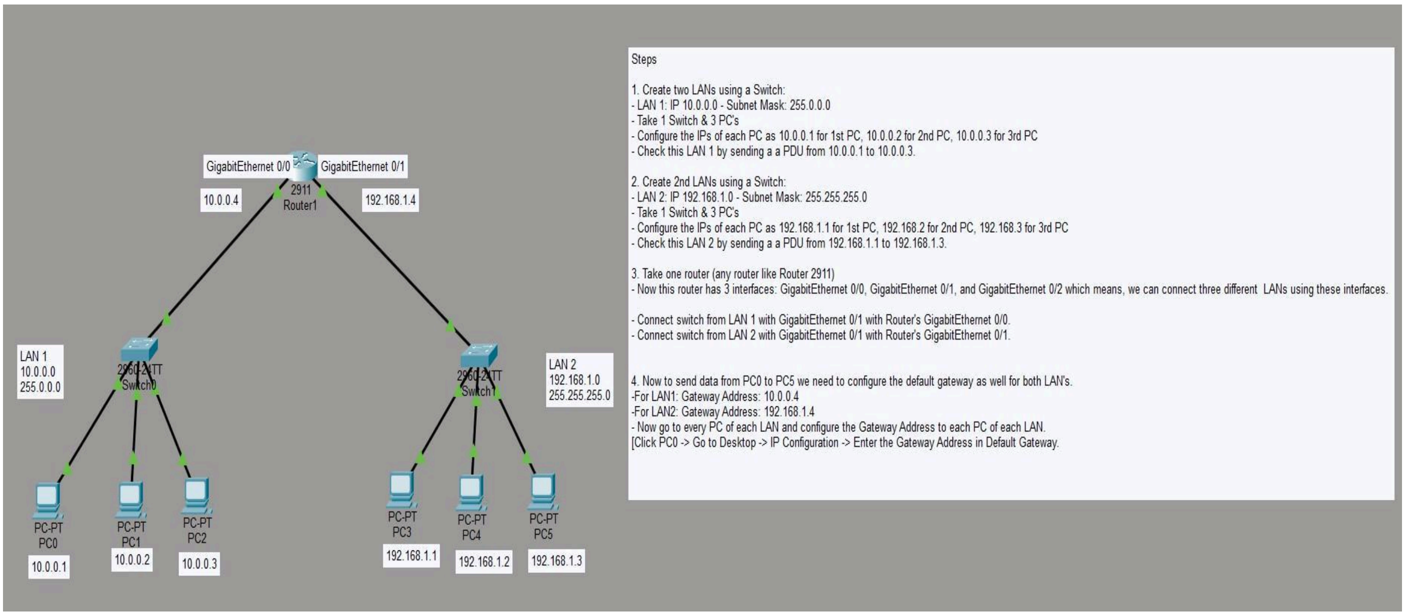


## LEARNING OUTCOMES:

After completing this experiment:

1. I understood the working principles of Hub, Switch, and Router by studying their functions, operating layers in the OSI model, and roles in establishing communication within and between computer networks.

2. I learnt about the configuration and implementation of Local Area Networks (LANs) using Cisco Packet Tracer by connecting multiple PCs with hubs and switches, assigning valid IPv4

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addresses and subnet masks, and verifying successful communication using the Ping command and Simple PDU simulation.

3. I developed practical knowledge of data forwarding mechanisms by observing that a Hub broadcasts data to all connected devices, a Switch forwards frames using its MAC Address Table, and a Router forwards packets between different networks using IP addresses and a Routing Table.

4. I acquired hands-on experience in configuring routers by assigning IP addresses to router interfaces, configuring default gateways for different LANs, connecting multiple networks, and enabling communication between hosts located on separate subnets through proper routing.

5. I enhanced the ability to analyze, compare, and troubleshoot networking devices by understanding the differences between Hub, Switch, and Router in terms of OSI layer, addressing method, forwarding technique, memory, collision and broadcast domains, duplex mode, security, performance, and practical applications.